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TITLE: Healthhub

SCHOOL: Chitkara University, Rajpura, Punjab

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ABSTRACT

HealthHub is a web-based healthcare platform that uses modern technology to make it easier to get medical care and make it work better. People who use traditional healthcare systems often have trouble making appointments, not being able to talk to their doctors very much, waiting a long time, and not being able to get help right away in an emergency. Because of these issues, the healthcare system is slow, disorganized, and hard to use. HealthHub's main goal is to bring patients, doctors, and donors together in one place so they can talk to each other. Users can sign up, log in, look at profiles of doctors, and make appointments based on when they are free and what they are good at. It also has chat and video consultation features that let patients talk to doctors in real time, so they don't have to go to the doctor's office.

One of the most important parts of the platform is the emergency blood request module. It lets people quickly ask for blood and get in touch with people who might be able to give it. This feature makes it easier to handle medical emergencies. The system also has dashboards that help users stay organized and open by letting them see their appointments, consultations, and request status.

React.js is used to build the front end of HealthHub, Node.js and Express.js are used to build the back end, and MongoDB/Appwrite is used to safely store data and log in. The platform is built to be safe, flexible, and responsive, so it works well on all devices. The platform also improves healthcare coordination as a whole by reducing the number of people who have to do things by hand and speeding up decision-making through digital workflows. It keeps users interested by giving them updates in real time, making it easy for them to get medical information safely, and making it easy for them to use. HealthHub brings together many different healthcare services into one system, making things easier, more reliable, and more efficient for both patients and healthcare providers.

HealthHub is a helpful and effective way to fix problems in today's healthcare system by making it easier to get medical care, speeding up communication, and cutting down on wait times. It helps run healthcare better and makes the healthcare ecosystem more connected and digital.

1.1 Problem Background

The healthcare industry is having a lot of problems because more people want medical care and the population is growing. Most traditional healthcare systems rely on tasks that people do by hand, like making appointments in person, going to the doctor, and not being able to talk to each other very much. This often leads to issues like long wait times, trouble getting an appointment with a doctor, and not being able to keep track of medical information. Patients, especially those who live far away, have a hard time getting the medical help and advice they need.

A lot of the time, patients and doctors can't talk to each other in real time, which makes it take longer to figure out what's wrong and treat it. Scheduling appointments by hand can cause problems like missed appointments and confusion for both patients and healthcare providers. It is also hard to get things like blood in a medical emergency because there is no central system.

Because of these limits, healthcare services don't work as well, patients aren't as happy, and medical care takes longer. There is a strong need for a digital solution that can make healthcare easier, help people talk to each other, and give them quick access to important medical services.

1.2 Objective of the Project

The main goal of this project is to create a smart and easy-to-use healthcare platform that will make it easier to get and manage medical services. The system's goal is to make it easier for patients, doctors, and donors to talk to each other by giving them a central place to do so. This will make it less likely that you will have to do things by hand. The project's main goal is to make important healthcare tasks easier. For instance, it will be easier to make online appointments, keep track of doctor profiles, and talk to each other in real time through chat and video calls. It also wants to add a special module for blood requests and donations so that people can get help quickly and reliably in an emergency.

There are also dashboards, real-time updates, and notifications on the platform to help people talk to each other and work together better. The system makes things more organized, cuts down on wait times, and overall improves healthcare services. The main goal is to create a healthcare system that is safe, scalable, and efficient, and that makes things easier, clearer, and better for both patients and healthcare providers.

1.3 Proposed Solution / System Overview

The proposed system is a web-based healthcare platform that aims to improve the efficiency, accessibility, and coordination of medical services. It has a single digital solution that makes it easy for patients, doctors, and donors to talk to each other through a simple interface. The platform gets rid of old-fashioned manual work and replaces it with a system that is organized and uses technology.

There are a number of user modules in the system, including Patient, Doctor, Donor, and Admin. Patients can sign up, log in, look at doctor profiles, make appointments, and talk to doctors over the phone or through chat. Doctors can manage their profiles, set their hours, and handle appointment requests. Donors can see and respond to requests for blood in an emergency. The admin is in charge of running the system, keeping track of user data, and making sure that everything works well on the platform.

Real-time updates, alerts, and dashboards make it easy for everyone to talk to each other. It also has a special emergency blood request module that makes it easy for people to quickly raise and respond to urgent medical needs.

There are a lot of important parts to the system. These include user authentication and management, an appointment booking system, a doctor-patient communication module (chat and video), an emergency blood request system, a notification service, and a central database. These parts work together to make sure that data is safe, services are delivered quickly, and healthcare is easy to use. The suggested system is a dependable and scalable way to make healthcare easier to get to, improve communication, and make sure that medical services are better managed.

1.4 Key Features or Functionality

There are a lot of things the system can do to make running healthcare services easier and more efficient. Patients can make, change, or cancel appointments online based on when the doctor is free. Users can quickly look at doctors' profiles, which list their specialties and years of experience, to help them make smart decisions.

Patients can chat and video call with doctors in real time on the platform. This means they don't have to go to the hospital in person. It also has a special feature for emergency blood requests that lets users make urgent requests and donors respond quickly. This is useful in emergencies.

HealthHub's dashboards are easy for both patients and doctors to use. Patients can see their appointment history, consultation history, and the status of their requests. Doctors can keep track of their schedules, how often they see patients, and when they are free. The system also tells you about changes to appointments, messages, and emergency activities.

Some of the other features that help keep user information private and safe are secure user authentication, profile management, and centralized data storage. Overall, these features make things more open, cut down on wait times, and improve the healthcare experience for everyone.

1.5 Technology / Method Used

The suggested system is built with modern web technologies and a modular full-stack architecture to make sure it can grow, work well, and be safe. The frontend is built with React.js, TypeScript, and Tailwind CSS. This makes it simple for patients, doctors, and donors to use. Node.js and Express.js are used to build the backend, which handles API requests, business logic, and system operations in a way that is quick and easy. MongoDB Atlas is the main database that keeps track of user data, doctor profiles, appointments, chat logs, and emergency blood requests in a neat way. JSON Web Tokens (JWT) are a way to check someone's identity and give them safe access. WebRTC and Socket.io are two technologies that let people talk to each other in real time, like in chat or video calls. This makes it easy for doctors and patients to talk to each other. RESTful APIs are also used to connect the front-end and back-end, which makes it easy for data and system operations to move smoothly. Cloud services like Vercel and Render host the platform, which makes it easy to access, reliable, and able to grow as needed.

<u>Layer</u>	<u>Technology Used</u>
Frontend	React.js, TypeScript, Tailwind CSS, Shadcn UI
Backend	Node.js, Express.js
Database	MongoDB Atlas
Real-Time Communication	WebRTC / Socket.io
Authentication	JWT (JSON Web Token)
Deployment	Vercel (Frontend), Render/Node server (Backend)

1.6 Expected Outcome & Benefits

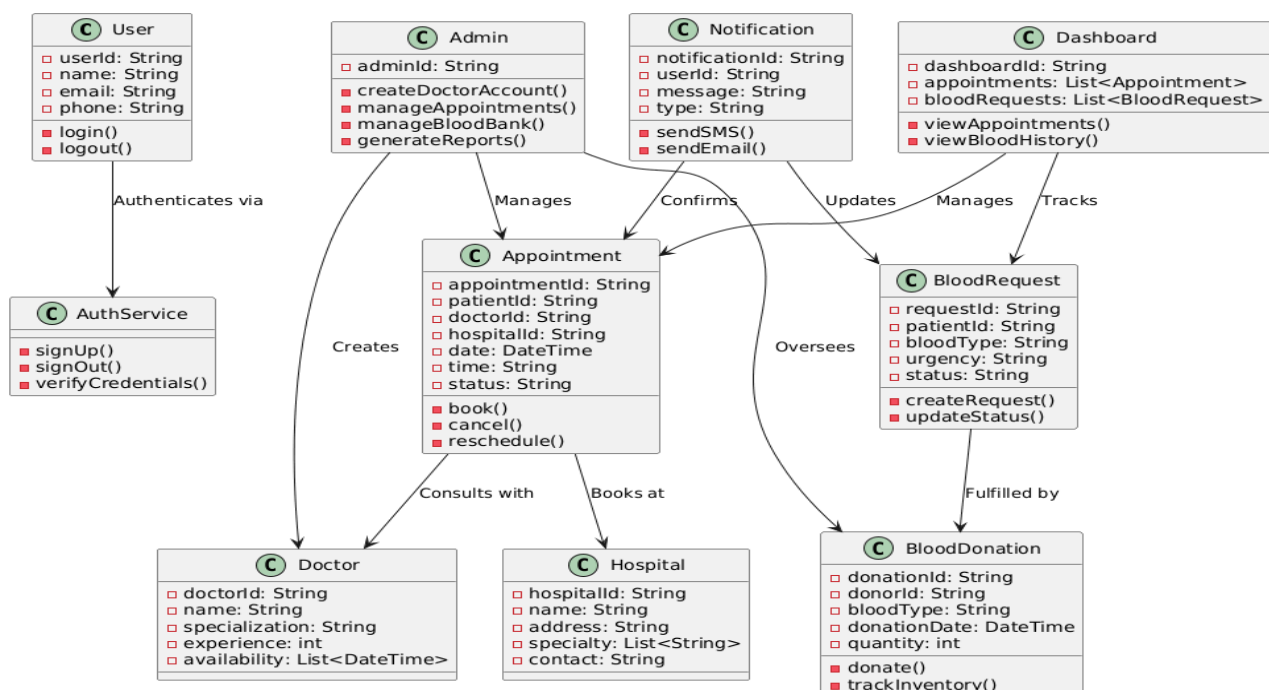
The proposed system is expected to make healthcare services much easier to get to, more efficient, and better organized by giving patients, doctors, and donors a central digital platform. It will have an online booking system that will make it easier to manage appointments, cut down on waiting times, and cut down on manual work. The patient will be able to easily find out about their doctor, talk to them in real time through chat and video consultation, and easily keep track of their medical activities. The emergency blood request feature will help handle emergencies better by quickly connecting people who need blood with people who can give it. Dedicated dashboards will help doctors keep track of their schedules, interactions with patients, and consultation records. The system is also in charge of securely handling data, sending notifications in real time, and improving communication, all of which lead to a better user experience and higher satisfaction. HealthHub is helping to change healthcare for the better in the long run by making services more open, trustworthy, and easy to use. This leads to better healthcare delivery and better overall health management.

1.7 Original Contribution Statement

This project is a unique integrated digital healthcare platform, which integrates the appointment management service, real time doctor-patient communication and emergency blood request service into a single system. Unlike existing solutions that provide limited or separate functionalities, HealthHub is a different approach. It combines multiple healthcare services in one location, such as doctor information, online consultation (chat and video) and emergency support. The platform concentrates on accessibility, efficiency and ease of use – but also on data security and role-based access control. The system is also designed with a modular and scalable architecture to allow for easy future extensions including AI-based health recommendations, advanced analytics and integration with wearable health devices. This makes HealthHub a valuable and novel contribution to the use of technology to reform healthcare services.

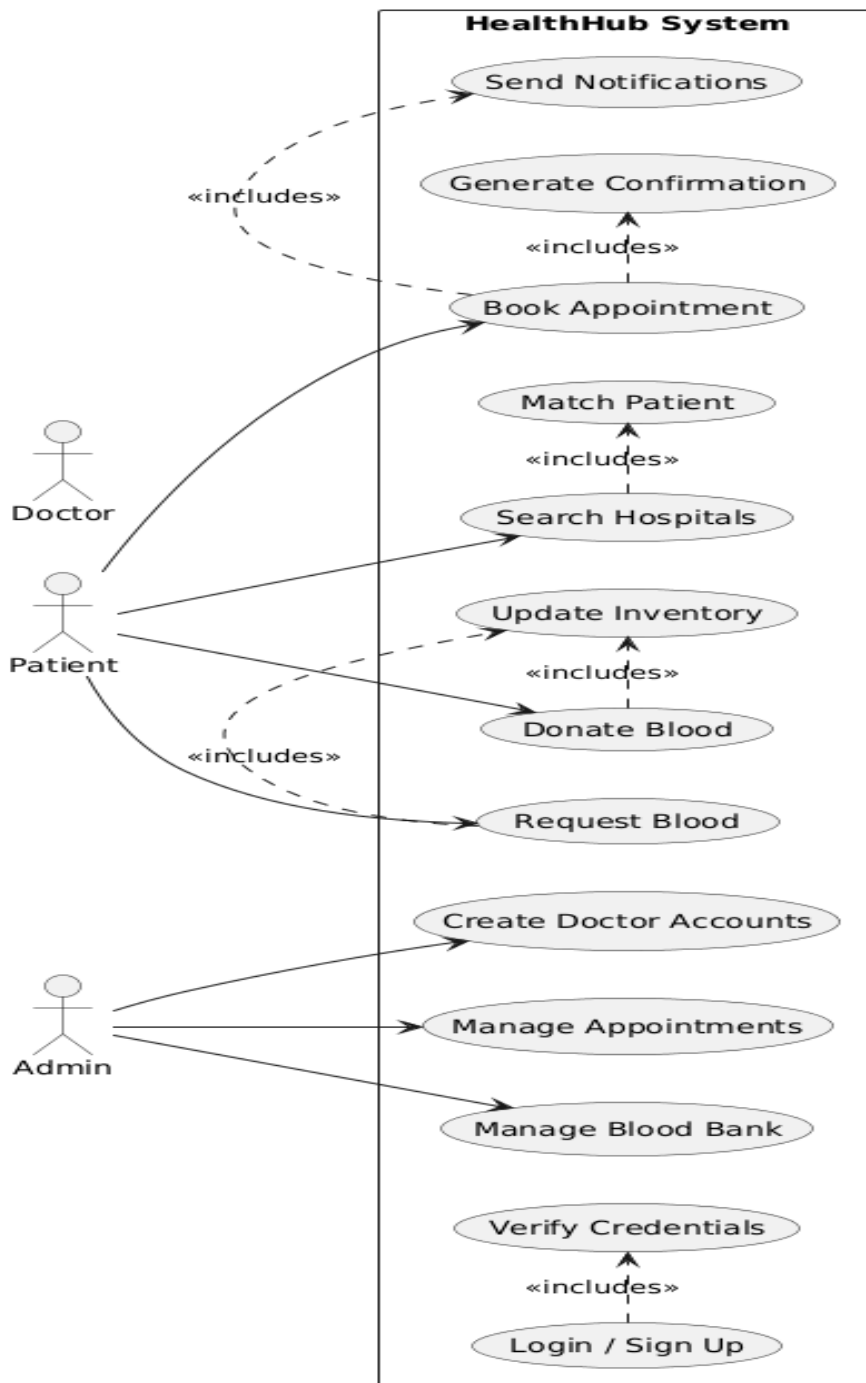
1.8 System Design

CLASS DIAGRAM:



The HealthHub – Class Diagram provides a structural view of the interaction between different classes in the healthcare system. It mainly focuses on the main objects like User, Dashboard, Appointment, Hospital, Doctor, BloodRequest, BloodDonation and service classes like AuthService, Notification, Admin. Every class has some attributes and methods that determine the class's behavior and associations within the healthcare platform. It allows scheduling appointments, blood bank operations, and hospital-doctor collaboration.

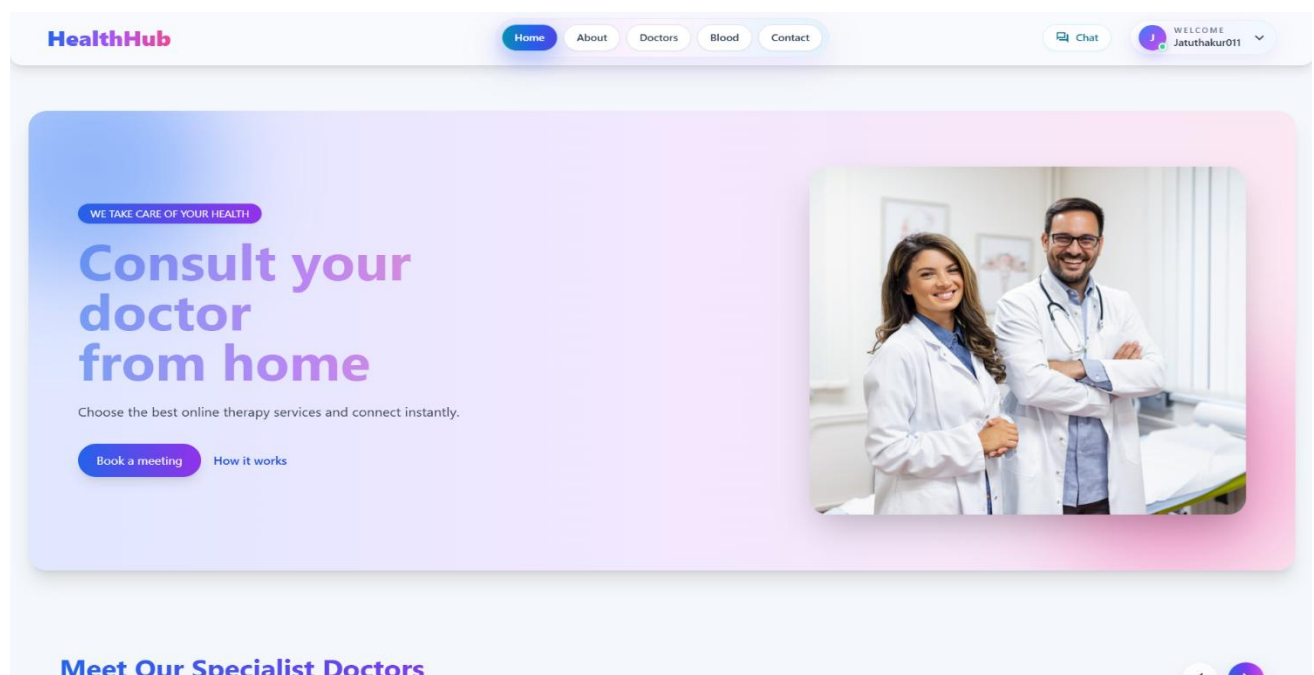
USECASE DIAGRAM:



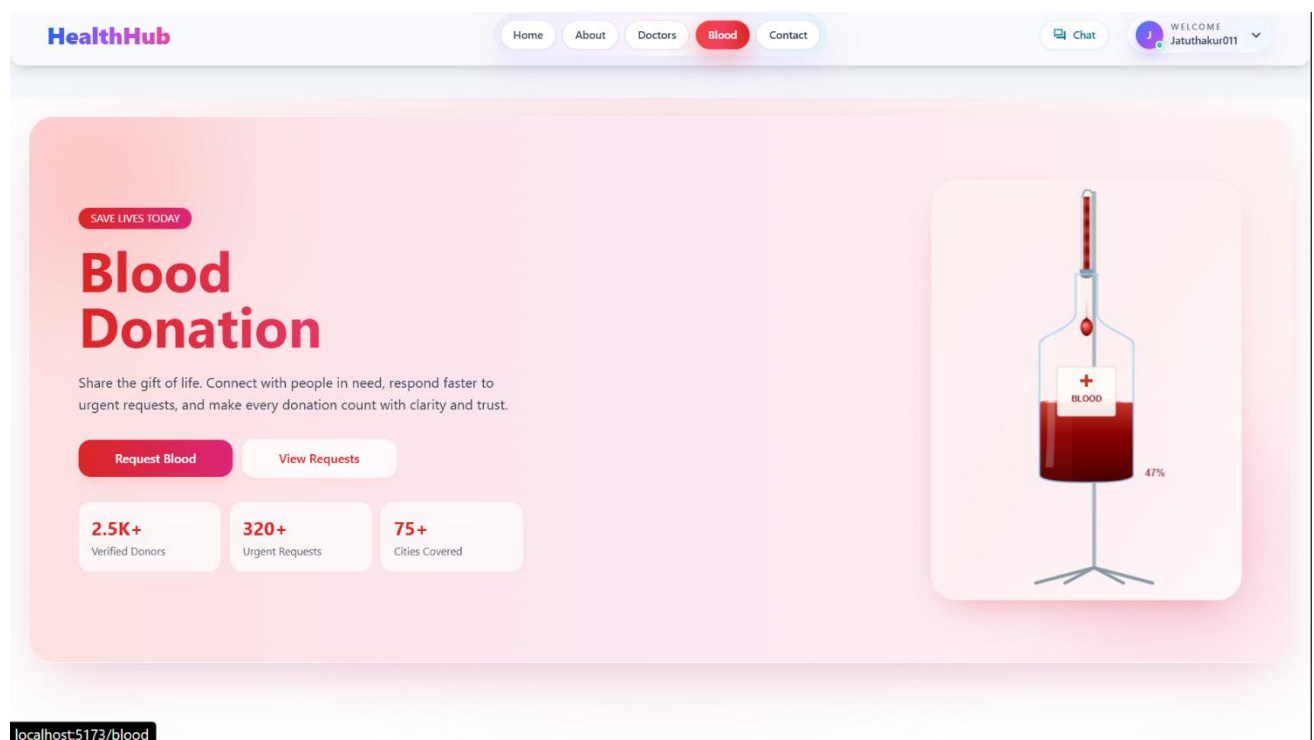
The use case diagram for HealthHub shows how different actors use the system to carry out different healthcare-related functions. The HealthHub platform involves four main actors: Patient, Doctor, Donor, and Admin, each responsible for certain interactions.

1.8 Results

Landing Page



Blood Donation Page:



localhost:5173/blood

DOCTOR PAGE:

HealthHub

[Home](#)[Chat 17](#)[Profile](#)[Video](#)[Bookings](#)[Logout](#)

ACTIVEINACTIVEWELCOME ram

DOCTOR COMMAND CENTERGood evening, Dr. ramSunday, 26 April 2026



Dr. ram
Specialist
PendingHours not setOnline

REQUESTS0

ACCEPTED0

RATING4.0

NEXT CONSULTATION

No upcoming visits. Accepted bookings appear here.

PENDING0

ACCEPTED0

RATING4.0

QUICK ACTIONS

Edit Profile
Photo, hours & bio

Requests
Review new cases

Bookings
Manage appointments

Consult Room
Chat and video care

TOTAL REQUESTS0
All time

ACCEPTED0
Confirmed visits

UPCOMING0
Ahead of schedule

AVG RATING4.0
1 review

PRACTICE PULSE
Today's Snapshot
Consultation readiness, trust score, and request flow at a glance.
80% SATISF.

BOOKING ACTIVITY
Recent Appointments

0 total

DOCTOR BOOKING PAGE:

HealthHub

[Home](#)[About](#)[Doctors](#)[Blood](#)[Contact](#)

ChatWELCOME Jatuthakur011

[← Back to Doctors](#)



HEALTHHUB VERIFIED

Dr. ram
Neurologist
★ 4.0/5 · 1 reviewExperiencedOnline

Available Now

Book Appointment Now

AVAILABILITY
Mon - Fri, 9:00 AM - 5:00 PM

LOCATION
Online

SPECIALTY
Neurologist

Services Offered

☒ In-person Consultation

☒ Follow-up Visit

☒ Online Video Session